

Exercises learning task 1

2026 – 2027

In Questions 2-1, 2-8 and 2-17, provide a reasonable description of the sample space for each of the random experiments. There can be more than one acceptable interpretation of each experiment. Describe any assumptions you make.

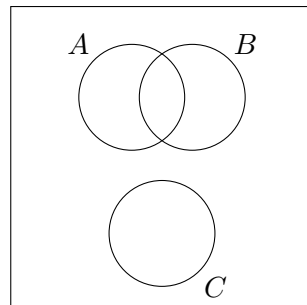
Exercise 2.1: Each of three machined parts is classified as either above or below the target specification for the part.

Exercise 2.8: The following two questions appear on an employee survey questionnaire. Each answer is chosen from the five-point scale 1 (never), 2, 3, 4, 5 (always).

- Is the corporation willing to listen to and fairly evaluate new ideas?
- How often are my coworkers important in my overall job performance?

Exercise 2.17: Calls are repeatedly placed to a busy phone line until a connection is achieved.

Exercise 2.20: Three events are shown on the Venn diagram in the following figure:



Reproduce the figure and shade the region that corresponds to each of the following events.

- | | |
|--------------------------|-----------------------------------|
| (a) A' | (b) $(A \cap B) \cup (A \cap B')$ |
| (c) $(A \cap B) \cup C$ | (d) $A \cap (B \cup C)'$ |
| (e) $(A \cap B)' \cup C$ | |

Exercise 2.29: The rise time of a reactor is measured in minutes (and fractions of minutes). Let the sample space be positive, real numbers. Define the events A and B as follows: $A = \{x \mid x < 65\}$ and $B = \{x \mid x > 45.5\}$.

Describe each of the following events:

- | | |
|----------------|----------------|
| (a) A' | (b) B' |
| (c) $A \cap B$ | (d) $A \cup B$ |

Exercise 2.31: A sample of two printed circuit boards is selected without replacement from a batch. Describe the (ordered) sample space for each of the following batches:

- (a) The batch contains 90 boards that are not defective, seven boards with minor defects, and three boards with major defects.

- (b) The batch contains 90 boards that are not defective, eight boards with minor defects, and one board with major defects.

Exercise 2.55: The sample space of a random experiment is $\{a, b, c, d, e\}$ with probabilities 0.2, 0.1, 0.2, 0.3, and 0.2, respectively. Let A denote the event $\{a, b, c\}$, and let B denote the event $\{c, d, e\}$. Determine the following:

- (a) $P(A)$ (b) $P(B)$ (c) $P(A')$
(d) $P(A \cup B)$ (e) $P(A \cap B)$

Exercise 2.74: If $P(A) = 0.4$, $P(B) = 0.2$, and $P(A \cap B) = 0.1$, determine the following probabilities:

- (a) $P(A')$ (b) $P(A \cup B)$ (c) $P(A' \cap B)$
(d) $P(A \cap B')$ (e) $P[(A \cup B)']$ (f) $P(A' \cup B)$

Exercise 2.77: Disks of polycarbonate plastic from a supplier are analyzed for scratch and shock resistance. The results from 100 disks are summarized as follows:

		Shock Resistance	
		High	Low
Scratch Resistance	High	70	10
	Low	15	5

- (a) If a disk is selected at random, what is the probability that its scratch resistance is high and its shock resistance is high?
- (b) If a disk is selected at random, what is the probability that its scratch resistance is high or its shock resistance is high?
- (c) Consider the event that a disk has high scratch resistance and the event that a disk has high shock resistance. Are these two events mutually exclusive?

Exercise 2.87: Disks of polycarbonate plastic from a supplier are analyzed for scratch and shock resistance. The results from 100 disks are summarized as follows:

		Shock Resistance	
		High	Low
Scratch Resistance	High	70	10
	Low	15	5

Let A denote the event that a disk has high shock resistance, and let B denote the event that a disk has high scratch resistance. Determine the following probabilities:

- (a) $P(A)$ (b) $P(B)$
(c) $P(A | B)$ (d) $P(B | A)$

Exercise 2.91: The following table summarizes the analysis of samples of galvanized steel for coating weight and surface roughness:

		Coating Weight	
		High	Low
Surface Roughness	High	36	16
	Low	114	34

- (a) If the coating weight of a sample is high, what is the probability that the surface roughness is high?
- (b) If the surface roughness of a sample is high, what is the probability that the coating weight is high?
- (c) If the surface roughness of a sample is low, what is the probability that the coating weight is low?

Exercise 2.106: Suppose that $P(A | B) = 0.25$, $P(A | B') = 0.35$, and $P(B) = 0.8$. What is $P(A)$?

Exercise 2.113: A batch of 30 injection-molded parts contains 6 parts that have suffered excessive shrinkage.

- (a) If two parts are selected at random, and without replacement, what is the probability that the second part selected is one with excessive shrinkage?
- (b) If three parts are selected at random, and without replacement, what is the probability that the third part selected is one with excessive shrinkage?